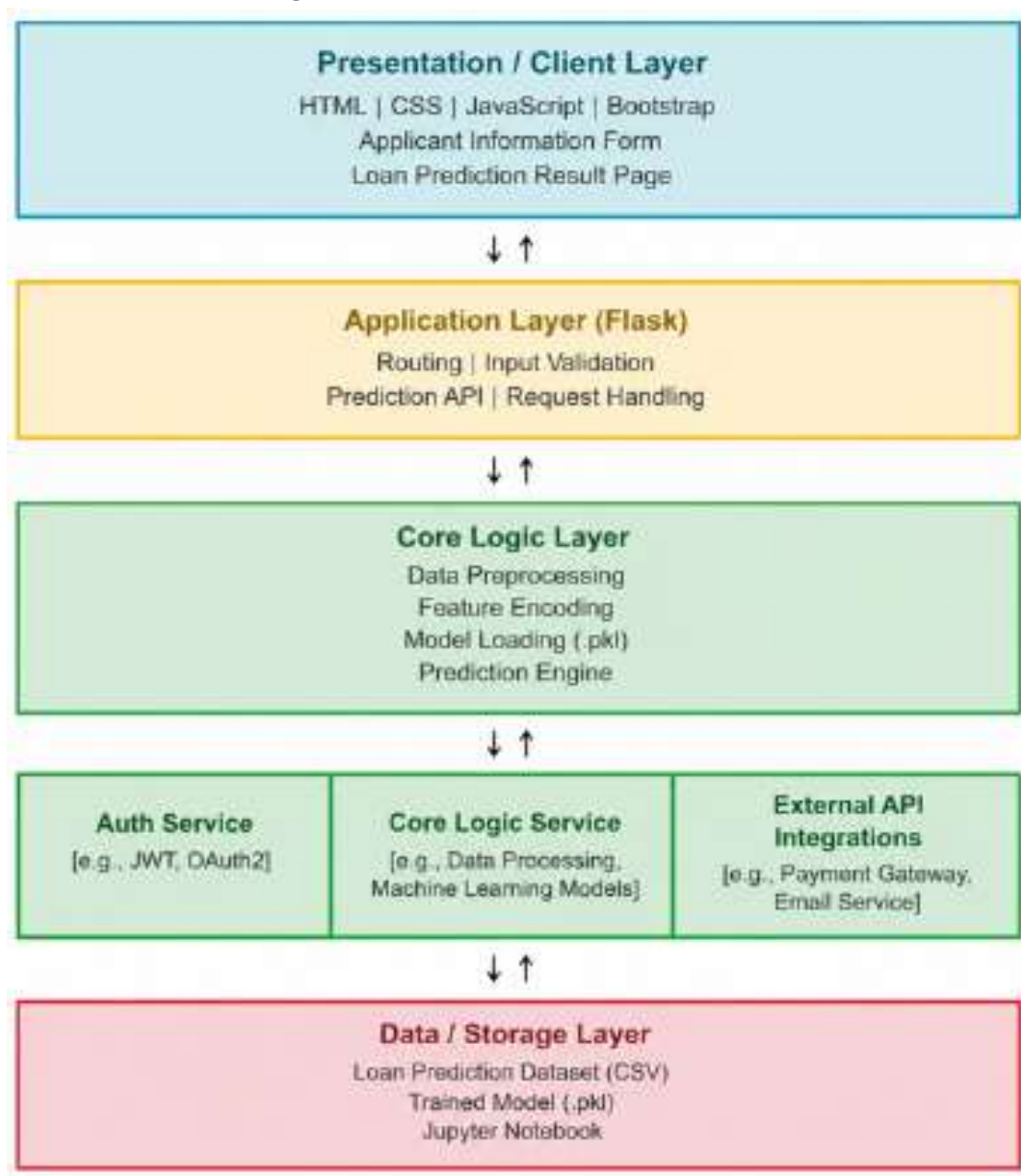


Solution Architecture

Date	26 June 2026
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Collects applicant details and displays loan prediction.</i>	<i>HTML, CSS, JavaScript, Bootstrap</i>
<i>Application Layer</i>	<i>Processes requests and connects UI with ML model.</i>	<i>Python, Flask</i>
<i>Core Logic Layer</i>	<i>Performs preprocessing, feature engineering, and prediction.</i>	<i>Pandas, NumPy</i>
<i>Machine Learning Layer</i>	<i>Predicts loan approval using trained models.</i>	<i>Decision Tree, Random Forest, KNN, XGBoost</i>
<i>Storage Layer</i>	<i>Stores dataset and trained model.</i>	<i>CSV Dataset, Pickle (.pkl)</i>
<i>Development Environment</i>	<i>Used to build and test the application.</i>	<i>VS Code, Jupyter Notebook, PyCharm</i>